

AIRCRAFT SCOUTS.

Their Value in Naval Warfare in Locating Submarines.

NOT GOOD AT SINKING THEM.

While They Are Clever at "Spotting" the Submerged Boats, It Is Only by Great Skill and Good Luck That a Bomb Can Be Dropped to Its Mark.

In some ways an aeroplane is more effective in locating a submarine than a destroyer or a patrol boat, and it may even be able to destroy the undersea vessel, says C. G. Grey, editor of the London Aeroplane.

There is a popular illusion, which is not exactly a delusion, that the observer in an aeroplane can see as far down into the water as a submarine is likely to dive. Up to a point this is true, for if the water is clear it is possible to see some twenty or thirty feet into it from a point vertically above, and if such water is only forty or fifty feet deep a submarine is not likely to dive beyond the visibility point.

It is quite another matter, however, where the water is a kind of leaden gray in color and where ten feet of water over the periscope is enough to hide a submarine as effectively as if she were twenty fathoms down. Therefore it is fairly obvious that the scouting air craft have to depend on surface vision quite as much as the patrol boats. Where the difference does come in is in the angle of vision from which that surface view is obtained.

For instance, a patrol boat might well miss seeing a submarine half a mile away, especially if the periscope were between the boat and the sun, so that the reflection of the sun off the water was blinding the lookout men. The aircraft observer, on the other hand, perched up aloft, would have a far better chance. The periscope, cutting through the water, makes a very distinct wake which spreads out on each side so that it makes a light but quite distinct "broad arrow" of foam.

When the periscope is above water the top of the conning tower is not far below, so that this also is visible unless there is a heavy breaking sea. And probably the bubbles from the exhausting air can also be seen.

Thus an escort of aeroplanes should have a far better chance of spotting the presence of submarines than could any escort of surface ships. At the same time it is well to remember that an escort of destroyers or some similar type of armed ship is necessary, because the aeroplane or airship of today is not capable of carrying guns big enough to sink submarines with certainty, and the art of bomb dropping from aircraft has not become sufficiently accurate to make the certainty of the bombs reaching their mark as great as is the certainty of a shell from a four inch gun doing so.

Therefore the aircraft by themselves can only hope to keep submarines submerged by dropping bombs at them, though they may sink one here and there by a combination of personal skill and good luck, whereas by acting as "spotters" for the highly trained gunners of destroyer flotillas they can co-operate very materially in the actual destruction of enemy submarines.

Where the naval air craft really do come in is in the immense surface of sea which they can cover in a very short time and their ability to take enemy submarines by surprise while lying on the surface "airing" themselves and charging up their batteries. A small fast scouting aeroplane does anything between eighty and a hundred miles an hour, and at a height of 7,000 or 8,000 feet it is fairly hard to see. On land one is generally warned of an aeroplane's arrival by the sound of its engine, but at sea the lapping of the water and the whistling of the wind through rigging or around superstructure drowns the sound of the aero engine till it is fairly close. Consequently a seaplane has more than a sporting chance of getting within a mile or so of a submarine before it is seen, and if it gets within that distance it should be able to make things very unpleasant for the submarine, even if it has not the luck to sink it outright.

Even at night in moderately fair weather, which is naturally the best for submarine operations, big seaplanes with searchlights could do much to keep submarines under water and so add to their troubles by preventing them from stopping on the surface long enough to air their interiors and to charge their storage batteries.—New York World.

Resistance of the Wind.
Tests on a model of the naval collie Neptune made in the wind tunnel of the Washington navy yard by Naval Constructor William McEntee show that if this vessel were steaming against a thirty mile wind at 14 knots an hour it would require about 770 horsepower to overcome the resistance of the wind. This is about 20 per cent of the power necessary to propel her through the water.

Very Formal.
"Are you on very friendly terms with your neighbor in the apartments?"
"Well, no. She's rather formal—always sends her card when she wishes to borrow flour, and if she wants both flour and sugar she sends two cards."—Washington Herald.

One must fight as an angel for freedom, but in freedom one must live as a saint.—Servian Proverb.

Chase the Ache, Kill the Pain.
Get busy. Act now. Use Sloan's Liniment for your rheumatic pains, toothache, neuralgia, sore muscles, stiff joints, sprains and strains. Better than anything you ever tried to soothe hurts, reduces swelling and inflammation. Cleanser than ointments or plasters, as it does not clog the pores or stain the skin. Easy to apply, it penetrates without rubbing. Always have a bottle of Sloan's Liniment in your medicine chest. At your druggist, 25c, 50c, \$1.00.

VAMPIRE OF THE OCEAN.

"Ironing" the Devilfish is a Made of Flirting With Death.

"Ironing" the devilfish, the largest of all the rays and one of the largest creatures of the sea, means flirting with death. The devilfish is also known as the ocean vampire. Its scientific name is Manta vampyrus, and all these names tell something of its habits. It gets its name devilfish from the fact that projecting from either side of the head is a hornlike appendage which in reality is a detached part of the pectoral fin or wing. The horns, sometimes three feet long and freely movable, are used to bring food to its mouth.

These fins have been known to circle above the bow of a boat, lifting the vessel out of the water and deluging the occupants of the boat with gallons of water. When they flop back to the surface they make a noise that can be heard far off. Exciting? Dangerous? The sport is both. The only thing that saves devilfish hunters at a time like this is preparedness. Every man must be at his station, all must obey the will of the harpooner, and if a bit of luck is thrown in the manta is killed.

The devilfish gets its name of manta, meaning blanket, from the pearl divers of the Caribbean sea, who are of the belief that it devours people after enveloping them in its enormous wings.

The fish has prodigious strength and has been known to tow a hundred ton vessel far out to sea. An authenticated story is told of a manta that towed eight boats, lashed together, for hours until finally the crews were compelled to cut the rope and let the devilfish escape.

Its vitality is wonderful. There are records of specimens escaping after having been harpooned, lanced and shot many times with rifles of heavy caliber. It fights even after the brain and heart have been pierced. Death comes only when the spinal cord, back of the brain, is severed.—New York Sun.

TRAVEL IN THE AIR.

The Safety That Goes With Transportation by Aeroplanes.

It is a new idea that the aeroplane is a safe means of transportation in safe hands, yet it is an idea that we must firmly get into the popular mind. The average citizen is still frightened at the prospect of leaving the ground and having no support except the air itself. Yet at the speed which we expect an aeroplane to maintain—seventy or eighty miles an hour—there is no means of transportation that is so safe.

The obstructions that cause accidents with trains and automobiles do not exist for flying craft. A locomotive has to follow a definite track, which may cause a frightful calamity. The aeroplane is not so circumscribed. Its tracks are wherever it wishes to go. There are no obstructions in the air—unless we regard "air pockets" as such—no bridges to cross, no mountains to penetrate, no signals to run by and no switches to be misplaced.

Or compare this new craft with the rough road of the automobile. The slightest mistake or even inattention on the part of the driver, going, say, forty or fifty miles an hour, may hurl the machine over a precipice or overturn it on an embankment. But such a lapse on the part of an aviator has no such deplorable results. There are no precipices to fall from and no obstructions to collide with.

The aeroplane is even safer than some forms of water travel. The motorboat, going at a speed of forty miles an hour or even slower, is a more risky form of transportation than an aeroplane. If such a boat strikes even a small obstruction, such as a floating plank, its side is punctured and it sinks in a few seconds. But there are no such dangers in the air.—Orville Wright in Harper's.

Marines.
Marines—soldiers serving on ship-board—date back to the year 1694. When an order in council, dated Oct. 16, authorized 1,200 soldiers to be raised and formed into a regiment. More regiments were later on formed, and in the latter years of the French wars they numbered 32,000. The marines are today a feature of every navy, and in most countries officers of the marines are equal in rank with those in the army and navy.—London Chronicle.

Slaves of the Gown.
Some women live in a state of perpetual preparation of wardrobe. They never seem to acquire one in any approach to completion, and they are always in the act of either planning or of obtaining some separate portion of one in such a way that it would seem as if they lived to and for no other end or purpose. The materials of which their gowns are made must be of the nature of Penelope's web.—Memphis Commercial Appeal.

Womanly Retort.
"Women are terribly inconsistent," protested Mr. Kewlible.
"John," replied his wife, "you know the poet says, 'Consistency's a jewel.' And you told me yourself that much jewelry is not good form."—Washington Star.

The First Question.
"Opportunity is at your door. Wants you to go with him."
"Has he a car?"—Louisville Courier Journal.

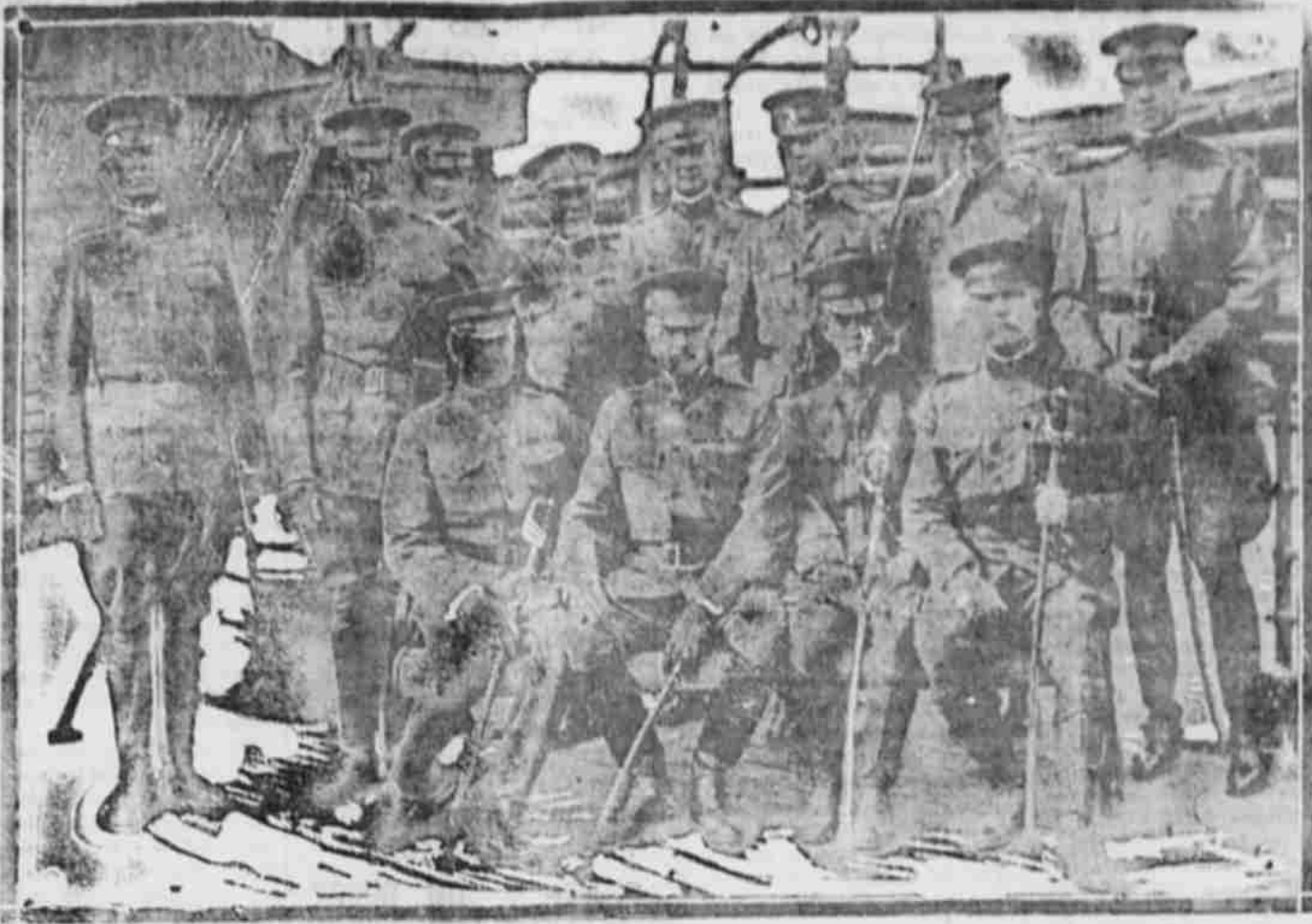
Ventriloquism.
Ventriloquism was first described in 725 B. C. Greeks ascribed it to the operation of demons.

Says He Abused Her.
Evora Smith today filed suit for divorce from Charles C. Smith, whom she claimed has abused her and made life miserable for her. They were married at Vancouver, Wash., July 26, 1913.

SON IS BORN.
Born, Saturday, June 30th, to the wife of Joseph C. Pettit, of West Linn, a son.

Medford to have a \$60,000 cold storage fruit warehouse.

First Photograph of General Pershing and Staff in England



A Champion at 15 Years



Bobby Jones, fifteen years old, of Atlanta, Ga., has won the southern golf championship. Young Jones sprang into prominence last year when he played in several tournaments in the east, and played many of the top notchers to a standstill. Young Jones is a member of the East Lake Golf club of Atlanta.

DAIRY WISDOM.

Don't sacrifice any better calf from a first class dairy cow.
Give each calf a chance to drink water at least twice each day.
Give the cows a good feeding of hay before turning into the fresh pasture.
Put the cows on the low ground where the coarse grass springs up and grows rankly at the start. If this grass is left uncropped it becomes tough and will not be eaten at all.
Calves can be raised perfectly on skim milk.

SUMMER SILAGE CROPS.

Provision Should Be Made to Tide Over Scant Pasture Period.

Dairy farmers who have not already made preparations for tiding over the period of insufficient pastures in late summer and fall should not forget that many cows went nearly dry last July and August, says a Pennsylvania Agricultural college bulletin. Such conditions may be avoided and incidentally the dairyman's bank account strengthened by a little forethought.

Maintaining the milk flow during the period of insufficient pasture may be accomplished by feeding additional grain, by feeding silage crops and by feeding silage.

Feeding silage crops in addition to the pasture secured by the cows helps to maintain the milk flow at moderate cost. The objection to this method is the labor and planning necessary in planting and cutting such crops at the proper time. Unless one has some silage from last year's feeding silage crops offer the most feasible feeding plan for the coming summer.

The best, cheapest and most practical way to tide over short pastures is to have an ample supply of corn silage. Plan to have enough left over from winter feeding to take care of the short pasture period. This involves less labor than silage crops, and the feed is handy and ready for use whenever needed.

The summer silo should be rather narrow in diameter so that the silage may be fed at the rate of three inches daily to prevent spoiling.

WHEATLESS RATIONS FOR CHICKENS BRINGS RESULTS THAT ARE EXCELLENT

With wheat so high, poultrymen will be interested to learn that on the United States Department of Agriculture experiment farm excellent egg-laying results were secured with a wheatless ration. Thirty Leghorn pullets to which this ration has been fed for 14 years produced an average of 147.3 eggs per hen for the pullet year. This compares favorably with egg yields secured on other rations containing wheat and therefore more expensive. This pen, moreover, during the first 16 weeks of its second year has averaged 23.5 eggs per hen, 17.5 eggs per hen being produced in March.

The same wheatless ration has been fed since last November to a pen of Buff Orpington pullets which have laid 33 eggs per hen in 26 weeks and hold the highest egg record of any of the large feeding pens of pullets at the farm this year. Two other pens, however, are less than 1 egg per hen behind this pen.

The ration used was as follows:
Scratch mixture—2 pounds cracked corn, 1 pound oats; Dry mash—3 pounds corn meal, 1 pound beef scrap.
The scratch mixture was fed sparingly so that the hens ate about equal parts of this mixture and of the dry mash. The total grain consumption of feed for the year was 52 pounds, of which 25 pounds was scratch mixture. Throughout the year it took 4.6 pounds of feed to produce a dozen eggs.

This scratch mixture, with wheat \$2.57, cracked corn \$1.35, and oats 70 cents per bushel, is 49 cents per hundred pounds cheaper than the regular mixture of equal parts cracked corn, wheat, and oats. Another mash which is slightly cheaper than this one, is made of 4 per cent each bran and middlings, 26 per cent beef scrap, and 66 per cent corn meal.

If the wheat is omitted from the ration it is very essential to feed a considerable proportion of beef scrap in the mash, but with present prices, beef scrap is one of the cheapest poultry feeds, considering its high protein content.

These experiments, the specialists say, prove that wheat is not essential in an egg-laying ration and that excellent results can be secured by using corn and oats as a scratch mixture provided this is fed with a good mash containing 25 per cent beef scrap.

ONE GIRL IS KILLED AND TWO OTHERS ARE WOUNDED BY ESCORT

LACONIA, N. H., July 2.—Miss Elizabeth Suess, assistant matron, and Miss Dorothy Davis, a teacher at the state school for feeble-minded here, were suffering today from wounds on the head and nervous shocks received when their companion, Miss Alice Black Richards, head teacher at the school, was murdered in a cottage on the shore of Lake Winnisquam last night. Morris P. Bradford, who had been with the women and gave himself up to the police, is held and will be given a hearing today.

According to the brief statements obtainable from Miss Suess and Miss Davis, the three women had accepted Bradford's invitation to take supper with him at the cottage, which is on the school grounds. There Bradford attacked them, they declared. He bound all three hand and foot, cut Miss Richards' throat, struck the other two over the head and cut their hair off. After he left one of the young women managed to free herself and released her companions, but Miss Richards had succumbed to her wounds.

Run-down? —Tired? —Weak?

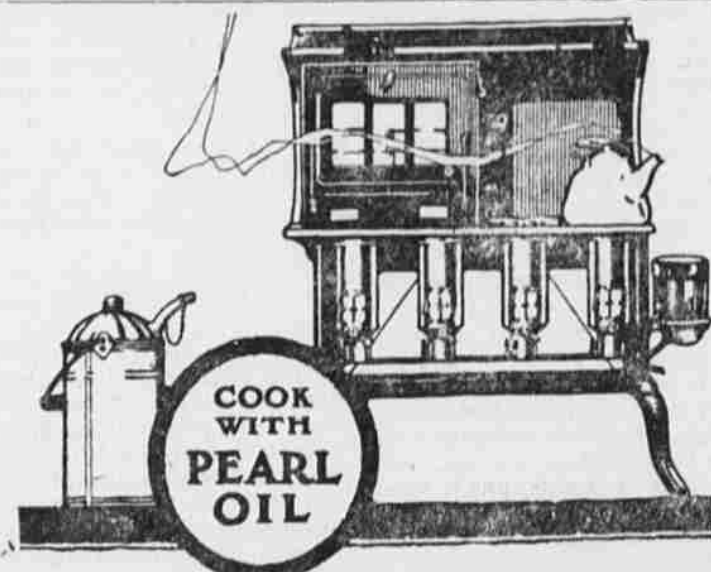
Every spring most people feel "all out of sorts"—their vitality is at a low ebb. Through the winter months shut up a great deal in heated houses, office, or factory, with little healthy exercise in the great outdoors—eating more than necessary—the blood becomes surcharged with poisons! The best spring medicine and tonic is one made of herbs and roots without alcohol—that was first discovered by Dr. Pierce—years ago. Made of Golden Seal root, blood root, with glycerine, it is called Dr. Pierce's Golden Medical Discovery. Ingredients on wrapper. It eliminates from the blood disease-breeding poisons. It makes the blood rich and pure, and furnishes a foundation for sound, physical health.

Sold by medicine dealers, in liquid or tablet form, or send 10 cents to Dr. Pierce's Invalids' Hotel, Buffalo, N. Y., for a large trial package of tablets.

WHEN "RUN-DOWN."
Salem, Oregon.—"As a spring tonic, to build up a weakened, run-down system, and to give one an appetite, I found Doctor Pierce's Golden Medical Discovery good. A friend had recommended it and I found it all that she claimed for it."—MRS. JAKE GUNDER, 1506 S. Bellevue Street.

A Clear Skin Your Birthright.

If your skin is not smooth, fresh and clear, do as others do and give it the remedy it needs to restore it to normal health and beauty. Try the soothing and healing treatment of Dr. Hobson's Eczema Ointment. The severest cases of eruptions, pimples, blotches and burning skin yield rapidly to this remarkable preparation. You'll be gratified with the rapidity with which your complexion regains its health and freshness. At your druggist, 50c.



ECONOMY

An oil cook-stove is cheaper to buy than a wood or coal stove and it's much cheaper to operate. Meals in a jiffy, and a cool kitchen in summer.

All the convenience of gas—economical for all the year 'round cooking. Bakes, broils, roasts, toasts. Steady, evenly-distributed heat, the best for cooking.

The long blue chimneys prevent all smoke and smell. In 1, 2, 3 and 4 burner sizes, with or without ovens. Also cabinet models. Ask your dealer today.

NEW PERFECTION OIL COOK-STOVE

FOR SALE BY

L. ADAMS DEPARTMENT STORE, HOGG BROS. FRANK BUSCH C. W. FRIEDRICH WILSON & COOKE

Oregon City, Ore.

Summer Hints

Do not allow garden vegetables to crowd each other in drills or rows. This is a bit of timely cultural advice offered to home gardeners by specialists of the United States Department of Agriculture. Unless each plant has adequate room for development and a chance to receive all the sunlight, air, moisture and plant food which it needs, the specialists point out, it will be stunted and its contribution to the food supply of the family therefore will be low in quantity and inferior in quality.

It is well enough to plant seed more thickly than the plants should grow, to allow for faulty germination and losses, it is explained, but the mistake of allowing the crops to grow as thickly as they come up must not be made. A good time to thin out surplus plants is a few weeks after the plants have come up, when they have become large enough to show which are most thrifty, and before they have begun to crowd seriously. Wherever possible, the thickest plants should be left.

In thinning lettuce and beets, some of the best rooted plants may be set out to fill up any spaces which may have occurred in the drills. Lettuce also may be permitted to grow fairly large before thinning, and the surplus plants eaten. The plants should be from 2 to 3 inches apart at the first thinning, and before the plants crowd a second thinning should be made which will leave the plants from 6 to 8 inches apart. The plants removed at the second thinning should be large enough for table use. Garden beets should be gradually thinned to 6 or 8 inches apart, the plants pulled being used for greens. For this purpose the whole plant is used. Young turnips may be used in the same way.

Radishes are usually not thinned until they are large enough to be eaten. Radish tops also may be cooked as greens or, if especially tender, may be used in salads. If radishes have been planted to mark drills of parsnips, carrots or other slow-growing vegetables, they should be removed before they crowd the permanent occupants of the drills.

Onions should be thinned before they reach a quarter-inch in diameter, and should be left about three inches apart.

Turnips, carrots, parsnips and beets usually are left 3 to 6 inches apart. If they are to be harvested while fairly small they may stand somewhat closer. Thin hills of muskmelons and cucumbers to 2, or at the most 5 plants. Only 2 or 3 pole beans should be left to each hill.

In all thinning operations choose, if possible, a time when the ground is mellow, so that little damage will be done to the plants that remain. If the ground surface about the plants to be thinned is hard, wet it thoroughly long enough before operations are begun to allow the moisture to soak to the ends of the roots.

Many persons complain about feeling old before they should. Like a weak link in a chain, a weak organ enfeebls the whole body. Overworked, weak or disordered kidneys lower vitality. A. W. Morgan, Angolia, La., writes: "I suffered with pains in the back. I am 43 years old, but I felt like a man of 90 years old. Since I took Foley Kidney Pills I feel like I did when I was 21." In 50c and \$1.00 sizes. Jones Drug Co.

UNION MEETING IS OFF.

The meeting of the employees of the woolen mills, scheduled for Friday evening at which time the organization of a union was to have been effected, has been indefinitely postponed.

MYSTERY SURROUNDS FIRE DISCOVERED AT ASTORIA LUMBER MILL

ASTORIA, Or., July 2.—A fire of unknown origin that threatened to destroy the tramway at the plant of the Hammond Lumber company here, together with docks and thousands of feet of lumber nearby, gained good headway and was burning briskly at 8:15 o'clock last night, when it was extinguished by the crew of the Harkins steamer Lurline, bound from Portland to Astoria, and which was near the mill at the time. Through the prompt action of the steamer's crew the blaze was put out.

The steamer Lurline had put into the mill dock and loaded lumber, and was swinging into the stream when Captain W. W. McCulley, the master, saw the dock and tramway blazing. He immediately sounded the fire whistle, blowing it for several minutes and in the meantime ordering the crew to get the fire hose on the steamer connected up.

When the steamer reached the dock again the men hurried ashore and soon had the flames extinguished with the water pumped from the steamer.

Safety First Cough and Cold.
"Oh, just a cough" today may become gripe or pneumonia tomorrow. Thousands die from neglected colds. Take Dr. King's New Discovery before your cough becomes chronic. A few doses check the cold by killing the germs. The healing balsam soothes the throat, loosens the phlegm and clear the air passages of secretions which provoke coughing. Contains mildly laxative ingredients which remove the waste that aggravates the cold. At your druggist, 50c, \$1.00.